
Visual Token Compression for Efficient Vision-Language Models

장한나
2129032
통계학과
hanna2129032@ewhain.net

Abstract

As large language models (LLMs) begin to process visual information as tokens alongside text, visual token compression has gained increasing attention for improving the efficiency of LLM training and inference. Prior work has explored token selection based on attention scores or feature diversity. In this report, several methods inspired by existing approaches are implemented and compared under a fixed compression budget, with each evaluated by accuracy on a held-out validation set. Building on insights from existing methods, a hybrid approach combining attention-based selection and diversity-based clustering is further explored, ensuring that selected tokens are both semantically important and mutually diverse. Additionally, an ablation study is conducted by combining various prior methods to better understand the contribution of each component. Through these experiments, the hybrid approach achieves the best accuracy of 83.33% on the given validation set.

1 Introduction

As large language models (LLMs) have been rapidly developed in recent years, numerous efforts have emerged to extend their capabilities to handle diverse modalities beyond text. Vision-Language Models (VLMs) are one such result, enabling the joint processing of both visual and textual information within a unified LLM framework. However, VLMs face a critical challenge: the excessive number of visual tokens. Since a single image is typically encoded into hundreds or even thousands of visual tokens — far exceeding the number of text tokens — this imbalance introduces significant inefficiency in both training and inference.

To address this issue, prior work has explored various token compression strategies. One line of research ((1), (2), (3)) selects visual tokens based on attention scores, either from the vision encoder or from internal LLM layers, under the assumption that highly attended tokens carry more relevant information. Another line of research ((4), (5)) argues that feature diversity is equally important, proposing methods that select tokens based on pairwise distances or similarities to ensure broad coverage of visual content.

In this report, token selection strategies based on the core ideas of representative prior works from both lines of research are implemented and evaluated under a fixed compression budget. Furthermore, based on the results of these implementations, a hybrid approach combining attention-based selection and diversity-based clustering is explored. Specifically, three different strategies for merging these two criteria are investigated, and the best-performing variant achieves an accuracy of 83.33%, suggesting that jointly considering attention and diversity leads to more effective token selection. Additional hybrid combinations are also examined as part of an ablation study.

The main contributions of this report are as follows:

- Implementation and comparison of token selection strategies based on the core ideas of representative prior works on visual token compression.
- Exploration of a hybrid method combining vision encoder attention scores and K-means clustering, which achieves the best accuracy among all evaluated methods.
- Investigation of additional hybrid strategies, including combinations of vision encoder attention and LLM-internal attention, providing further insights into the complementary roles of different compression signals.

2 Related Work

In this section, prior works on visual token compression are reviewed, with a focus on the two main approaches explored in this report.

2.1 Attention-Based Methods

A number of prior works have attempted to select visual tokens based on attention scores. FastV (2) utilizes attention scores from within the LLM, specifically measuring how much each text token attends to each visual token, and prunes visual tokens that receive low attention. VisionZip (3) and FasterVLM (1), on the other hand, leverage attention scores from the vision encoder, particularly the attention of the CLS token toward each patch token, as a proxy for token importance. These methods establish a straightforward yet effective criterion for token selection: tokens that receive higher attention scores are considered more informative and are thus retained.

In this report, token selection strategies based on the core ideas of both approaches — CLS token attention from the vision encoder and text-to-visual attention from the LLM — are implemented under the given model and fixed compression budget. The results are reported in the Experiments section.

2.2 Diversity-Based Methods

Following the success of attention-based methods, several researchers argued that selecting tokens solely based on attention scores tends to produce redundant token subsets, as highly attended tokens often capture similar visual information. To address this limitation, DivPrune (4) and CDPPruner (5) propose selecting tokens that maximize the diversity of the selected subset, based on pairwise distances between token features or conditioned on the input instruction. Similarly, K-means clustering ((6), (7)) has been explored as a way to partition the token space and select representative tokens from each cluster, leveraging the inherent diversity-preserving property of clustering.

In this report, token selection strategies based on greedy diversity selection and K-means clustering are implemented under the given model and fixed compression budget. The results are reported in the Experiments section.

3 Method

Through the implementation of prior approaches, it is observed that attention-based and diversity-based methods each capture complementary aspects of token importance. This motivates the exploration of a hybrid approach that combines both criteria. Based on the implementation results and the relative stability of each method, a combination of CLS attention scores and K-means clustering is selected as the basis for the hybrid method. Specifically, three different strategies are investigated to determine the most effective way to integrate these two criteria.

3.1 Attention-First Hybrid

In this strategy, the top-N tokens are first selected according to the CLS token attention scores of the vision encoder, and K-means clustering is then applied to identify 64 representative tokens from this attention-filtered subset. This approach is motivated by the assumption that enforcing diversity among already-important tokens — rather than across all tokens — leads to better performance, as attention is treated as the primary criterion.

3.2 Clustering-First Hybrid

In this strategy, clustering is applied first, followed by attention-based selection, under the assumption that diversity should be prioritized over attention. Two variants are explored within this strategy.

In Method I, K-means clustering is applied to obtain N clusters, and the representative token of each cluster is selected as the centroid-nearest token. The top-64 tokens are then chosen from these N representatives based on their CLS attention scores.

In Method II, K-means clustering is applied to obtain exactly 64 clusters, and within each cluster, the token with the highest CLS attention score is selected as the representative.

Among all investigated strategies, Method I of the Clustering-First Hybrid approach achieves the best accuracy, and this strategy is adopted in the submitted main.ipynb. Overall, the hybrid methods achieve performance comparable to or better than single-criterion methods, with Method I in particular surpassing both attention-based and diversity-based methods alone. This result suggests that prioritizing diversity in the initial selection stage and subsequently refining the selection based on attention scores is an effective strategy. Furthermore, the performance gap between Method I and Method II indicates that using a larger number of clusters than the target token count allows for a more flexible and effective integration of attention-based refinement.

4 Experiments

In this section, the experimental results of all implemented methods are presented and analyzed. All methods are evaluated under a fixed compression budget of 64 visual tokens on the provided validation set of 30 samples. For hybrid methods, n_{init} is tuned over 3, 5, and it is found that $n_{\text{init}} = 3$ consistently yields better or equivalent performance across all configurations. Therefore, only results with $n_{\text{init}} = 3$ are reported in the following sections. Additionally, detailed hyperparameter tuning results for each method are provided in Appendix A.

4.1 Attention-Based Methods

Two attention-based methods are evaluated: CLS token attention from the vision encoder and text-to-visual attention from the LLM. For the CLS token attention method, tokens are selected based on the CLS token’s attention scores from the vision encoder. Attention scores are extracted from individual layers as well as averaged across all layers, in order to identify the most effective layer configuration. It is found that the last layer yields the best performance, achieving 80% accuracy. For the LLM attention method, tokens are selected based on text-to-visual attention scores from within the LLM, and the same layer-wise analysis is conducted. It is found that averaging attention scores across all layers works best for this method, also achieving 80% accuracy. Through the attention-based experiments, it is confirmed that attention scores alone are sufficient to achieve competitive performance.

4.2 Diversity-Based Methods

Two diversity-based methods are also evaluated: greedy diversity selection and K-means clustering. For greedy diversity selection, tokens are selected iteratively based on their distance from the already-selected token subset, either measured by L2 distance or cosine similarity, until 64 tokens are chosen. Through a grid search over the first token selection strategy and distance metric, it is found that randomly selecting the first token and using cosine similarity as the distance metric yields the best performance, achieving 76.67% accuracy. For K-means clustering, 64 clusters are first constructed from the visual token features, and the representative token of each cluster is selected based on either L2 distance or cosine similarity to the cluster centroid. Through hyperparameter tuning, it is found that using cosine similarity as the distance metric with n_{init} set to 3 or 5 yields the best performance, also achieving 76.67% accuracy. Through the diversity-based experiments, it is found that diversity-based methods generally yield lower accuracy than attention-based methods. Between the two diversity-based methods, K-means clustering demonstrates more consistently higher performance across multiple configurations, although both methods achieve the same best accuracy.

4.3 Attention-First Hybrid

For the Attention-First Hybrid method, the top-N tokens are first selected based on CLS attention scores, followed by K-means clustering to identify 64 representative tokens. Based on the findings in Sections 4.1 and 4.2, cosine similarity is adopted as the distance metric. The value of N is tuned over the range of 100 to 300, and it is found that $N = 100$ and $N = 250$ yield the best performance, both achieving 80% accuracy. Although the Attention-First Hybrid method achieves 80% accuracy, this is comparable to the performance of attention-based methods alone, suggesting that attention and diversity are not effectively integrated in this strategy.

4.4 Clustering-First Hybrid

For the Clustering-First Hybrid method, both Method I and Method II are evaluated. For Method I, which first constructs N clusters and then selects the top-64 representative tokens based on CLS attention scores, cosine similarity is adopted as the distance metric and the number of clusters is tuned over the range of 100 to 300. It is found that 250 clusters yield the best performance, achieving 83.33% accuracy. For Method II, which constructs 64 clusters and selects representative tokens based on CLS attention scores, an accuracy of 73.33% is achieved. The Clustering-First Hybrid method achieves the highest accuracy of 83.33% among all evaluated methods, suggesting that applying diversity-based clustering before attention-based refinement is a more effective strategy. This result highlights the importance of ensuring sufficient diversity in the initial token selection stage.

As shown in Table 1, Method I of the Clustering-First Hybrid approach achieves the highest accuracy among all evaluated methods.

Table 1: Comparison of all evaluated methods under a fixed compression budget of 64 visual tokens. The best accuracy for each method is reported.

Category	Method	Accuracy (%)
Baseline	Random Sampling	70.00
Attention-Based	CLS Attention (last layer)	80.00
	LLM Attention (layer average)	80.00
Diversity-Based	Greedy Diversity (random + cosine)	76.67
	K-means Clustering (cosine, n_init=3)	76.67
Hybrid	Attention-First Hybrid (N=100, 250)	80.00
	Clustering-First Hybrid Method I (N=250)	83.33
	Clustering-First Hybrid Method II	73.33

5 Ablation Study

In this section, various hybrid combinations of existing methods are explored in order to identify more effective token selection strategies and analyze the contribution of each component. Additionally, detailed hyperparameter tuning results for each method in this section, as well as those in Section 4, are provided in Appendix A.

5.1 CLS attention and Greedy diversity method

For this method, the top-N tokens are first selected based on CLS attention scores from the vision encoder, and the final 64 tokens are then selected iteratively based on pairwise distances between tokens. Cosine similarity is adopted as the distance metric, and the first token is selected randomly. It is found that $N = 100$ yields the best performance, achieving 80% accuracy. Comparing this result with the Clustering-First Hybrid method, it is suggested that K-means clustering serves as a more effective diversity-enforcing component than greedy diversity selection when combined with CLS attention scores.

5.2 CLS attention and LLM attention

For this method, 64 tokens are selected based on a weighted sum of the normalized CLS attention scores and LLM attention scores. Based on the findings in Section 4, the last layer of the vision encoder is utilized for CLS attention, and layer-averaged attention scores are utilized for LLM attention. The weighting factor α , which controls the relative contribution of the two attention scores, is tuned over the range of 0.0 to 1.0. It is found that $\alpha = 0.6$ yields the best performance, achieving 83.33% accuracy. This result suggests that combining two complementary attention signals — one from the vision encoder and one from the LLM — can achieve competitive performance comparable to the hybrid method combining attention and diversity-based selection.

5.3 LLM attention and K-means clustering

For this method, the Clustering-First Hybrid strategy (Method I) is applied using LLM attention scores instead of CLS attention scores. The number of clusters is tuned over the range of 100 to 300, and it is found that $N = 150$ and $N = 250$ yield the best performance, achieving 76.67% accuracy. This result is lower than that of the corresponding CLS attention-based variant in Section 4.4, suggesting that K-means clustering works more effectively when combined with CLS attention scores from the vision encoder than with text-to-visual attention scores from the LLM.

5.4 CLS attention, LLM attention and K-means clustering

For this method, the Clustering-First Hybrid strategy (Method I) is applied using a weighted sum of CLS attention and LLM attention scores, instead of CLS attention scores alone. Specifically, N clusters are first constructed, and the top-64 representative tokens are then selected based on the weighted sum of the two attention scores. The weighting factor α is fixed at 0.7, based on the findings in Section 5.2, and the number of clusters N is tuned over the range of 100 to 300. It is found that $N = 200$ or $N = 250$ yields the best performance, achieving 83.33% accuracy. Compared to the result of Section 5.3, where only LLM attention was used, the addition of CLS attention leads to a notable performance improvement, suggesting that CLS attention and K-means clustering operate in a complementary manner.

To summarize the ablation study, when combining attention and diversity signals, the combination of CLS attention and K-means clustering consistently achieves the best performance among all investigated strategies. When diversity-based methods are not considered, combining two complementary attention signals — CLS attention and LLM attention — is found to be the most effective alternative, achieving competitive performance comparable to the best hybrid method. A summary comparison of all ablation methods is provided in Table 2.

Table 2: Comparison of ablation study methods. The best accuracy for each method is reported. $\dagger$ denotes the best-performing method from Section 4.

Method	Best Configuration	Accuracy (%)
Clustering-First Hybrid Method I $\dagger$	$N=250$	83.33
CLS Attention + Greedy Diversity	$N=100$, random first token	80.00
CLS Attention + LLM Attention	$\alpha=0.6$ or 0.7	83.33
LLM Attention + K-means	$N=150$ or 250	76.67
CLS + LLM Attention + K-means	$N=200$ or 250 , $\alpha=0.7$	83.33

6 Conclusion

In this report, various visual token compression methods are explored and compared under a fixed compression budget of 64 visual tokens. Through extensive experiments, it is found that hybrid methods combining attention-based selection and diversity-based clustering almost always outperform methods that rely on either criterion alone. Specifically, the Clustering-First Hybrid approach, which prioritizes diversity by constructing clusters before applying attention-based refinement, achieves the best accuracy of 83.33% among all evaluated methods. This result implies that ensuring sufficient

feature diversity as a foundation, before applying attention-based refinement, is a more effective strategy for visual token selection. In other words, diversity should serve as the basis upon which attention-based selection is built, rather than the other way around. Furthermore, the ablation study reveals that CLS attention from the vision encoder and K-means clustering form the most effective combination, suggesting that these two components operate in a highly complementary manner. One notable finding from the ablation study is that combining CLS attention and LLM attention — without any diversity-based component — also achieves competitive performance of 83.33%. While this result is encouraging, the underlying reason why two attention signals alone can match the performance of attention-diversity hybrids remains insufficiently analyzed. A more thorough investigation into the complementary roles of different attention sources is left as future work.

References

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A Hyperparameter Tuning Results

This appendix presents the detailed hyperparameter tuning results for each method evaluated in Sections 4 and 5. For each method, all explored configurations and their corresponding accuracy on the validation set are reported.

A.1 CLS Attention

CLS token attention scores from the vision encoder are used to select visual tokens, and the attention layer to reference is tuned. Table 3 presents the accuracy for each layer configuration.

Table 3: Accuracy of CLS Attention method with varying layers.

Layer	Accuracy (%)
0	66.67
11	70.00
20	73.33
23 (last)	80.00
Average	70.00

A.2 LLM Attention

Text-to-visual attention scores from within the LLM are used to select visual tokens, and the attention layer to reference is tuned. Table 4 presents the accuracy for each layer configuration.

Table 4: Accuracy of LLM Attention method with varying layers.

Layer	Accuracy (%)
8	76.67
16	70.00
24	76.67
31 (last)	63.33
Average	80.00

A.3 Greedy Diversity

Tokens are selected iteratively based on pairwise distances, and the first token selection strategy and distance metric are tuned. Table 5 presents the accuracy for each configuration.

Table 5: Accuracy of Greedy Diversity method with varying first token selection strategy and distance metric.

First Token	Distance Metric	Accuracy (%)
Random	L2	70.00
	Cosine	76.67
Fixed	L2	70.00
	Cosine	73.33

A.4 K-means Clustering

Visual tokens are partitioned into 64 clusters, and the distance metric and n_init are tuned. Table 6 presents the accuracy for each configuration.

Table 6: Accuracy of K-means Clustering method with varying distance metric and n_init.

Distance Metric	n_init	Accuracy (%)
L2	1	63.33
	3	73.33
	5	73.33
Cosine	1	73.33
	3	76.67
	5	76.67

A.5 Attention-First Hybrid

The top-N tokens are first selected based on CLS attention scores, and K-means clustering is then applied to identify 64 representative tokens. The value of N is tuned. Table 7 presents the accuracy for each configuration.

Table 7: Accuracy of Attention-First Hybrid method with varying Top-N values. Cosine similarity is used as the distance metric and n_init is fixed at 3.

Top-N	Accuracy (%)
100	80.00
150	76.67
200	76.67
250	80.00
300	76.67

A.6 Clustering-First Hybrid

K-means clustering is first applied, followed by attention-based selection. Two variants are evaluated. Table 8 presents the accuracy for each configuration.

Table 8: Accuracy of Clustering-First Hybrid methods. For Method I, cosine similarity is used as the distance metric and n_init is fixed at 3. Method II uses 64 clusters with n_init fixed at 3.

Method	Number of Clusters (N)	Accuracy (%)
Method I	100	73.33
	150	76.67
	200	80.00
	250	83.33
	300	76.67
Method II	64 (fixed)	73.33

A.7 CLS Attention + Greedy Diversity

The top-N tokens are first selected based on CLS attention scores, and greedy diversity selection is then applied. The value of N and the first token selection strategy are tuned. Table 9 presents the accuracy for each configuration.

Table 9: Accuracy of CLS Attention + Greedy Diversity method with varying Top-N and first token selection strategy. Cosine similarity is used as the distance metric.

First Token	Top-N	Accuracy (%)
Fixed (top-1)	100	76.67
	150	70.00
	200	70.00
	250	70.00
	300	70.00
Random	100	80.00
	150	73.33
	200	73.33
	250	70.00
	300	70.00

A.8 CLS Attention + LLM Attention

CLS attention and LLM attention scores are normalized and combined via a weighted sum, and the weighting factor alpha is tuned. Table 10 presents the accuracy for each configuration.

Table 10: Accuracy of CLS Attention + LLM Attention method with varying alpha. CLS last layer and LLM layer average are used.

Alpha (CLS weight)	Accuracy (%)
0.0	80.00
0.1	76.67
0.2	73.33
0.3	70.00
0.4	73.33
0.5	80.00
0.6	83.33
0.7	83.33
0.8	80.00
0.9	80.00
1.0	80.00

A.9 LLM Attention + K-means

The Clustering-First Hybrid strategy (Method I) is applied using LLM attention scores, and the number of clusters is tuned. Table 11 presents the accuracy for each configuration.

Table 11: Accuracy of LLM Attention + K-means method with varying number of clusters. Cosine similarity is used as the distance metric and n_init is fixed at 3.

Number of Clusters (N)	Accuracy (%)
100	70.00
150	76.67
200	70.00
250	76.67
300	73.33

A.10 CLS + LLM Attention + K-means

The Clustering-First Hybrid strategy (Method I) is applied using a weighted sum of CLS and LLM attention scores, with alpha fixed at 0.7. The number of clusters is tuned. Table 12 presents the accuracy for each configuration.

Table 12: Accuracy of CLS + LLM Attention + K-means method with varying number of clusters. Alpha is fixed at 0.7, cosine similarity is used as the distance metric, and n_init is fixed at 3.

Number of Clusters (N)	Accuracy (%)
100	70.00
150	80.00
200	83.33
250	83.33
300	80.00